

Tutorial 01

Student Worksheet: Introduction to Software Architecture

Faculty of Computing – Sabaragamuwa University of Sri Lanka

Course: Software Architecture

Session: Introduction to Software Architecture

Duration: 60–75 minutes

Part 1 – Understanding the Concept

1. In your own words, describe what software architecture means.

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2. Why is software architecture important when developing large systems? List three reasons.

1.
2.
3.

3. Imagine a team starts building a large system without any architectural design. What problems might occur?

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Part 2 – Stakeholders

4. Who are stakeholders in a software system?

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5. Consider a mobile banking application. Identify four stakeholders and their concerns.

Stakeholder	Concern / Interest
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Part 3 – The 4+1 Architectural View Model

Match the views with the correct description.

View	Description
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Logical View

Process View

Development View

Physical View

Descriptions:

- a) Focuses on system functionality and objects
- b) Focuses on system performance and runtime behavior
- c) Focuses on hardware and deployment environment
- d) Focuses on source code organization and modules

Part 4 – Practical System Analysis

Consider an Online Food Delivery System where users can browse restaurants, order food, make payments, and track delivery.

6. Logical View – List four main objects/classes.

- 1.
- 2.
- 3.

4.

7. Process View – What runtime behaviors must the system handle?

1.

2.

8. Development View – What software components might exist?

1.

2.

3.

9. Physical View – Where might the system run?

1.

2.

3.

Part 5 – Scenario (The +1 View)

Write a simple user scenario for the system (User → Action → System Response).

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Part 6 – Reflection

10. Why is it useful to analyze software systems from multiple architectural views instead of just one?

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Optional Challenge Activity

Choose a real-world system (Uber, Instagram, Netflix, Online Library System) and identify the following views.

Architecture View

Your Answer

Logical View

Process View

Development View

Physical View